

Calculus 2a

Chapter 6 & 7

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Chapter 6 :- [Integration]

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6.1

* Anti-derivative :-

$f'(x)$ ~~is~~ $f(x)$ is ^{بزرگ} ~~big~~ ^{کوچک} ~~small~~
if the derivative is $(2x)$ what is the function

$$F(x) = x^2 \text{ or } F(x) = x^2 + 1, F(x) = x^2 + C$$

Anti-derivative

$$\Rightarrow F'(x) = 2x = \underset{\text{small}}{f(x)}$$

Def :- A function $F(x)$ is called an anti-derivative of $f(x)$ if $F'(x) = f(x)$

In case we write $\int f(x).dx = F(x) + C$

i.e. the Integral of $(f(x)) =$ The general Anti-deriv.

Ex. Find the general :-

1. $f(x) = 3x^2$

$$F(x) = x^3 + C$$

$$\text{or } \int 3x^2 dx = x^3 + C$$

← $\int x^2 dx = \frac{x^3}{3} + C$

← $\int x^3 dx = \frac{x^4}{4} + C$

2. $f(x) = \frac{1}{x} + \cos x$

$$F(x) = \ln |x| + \sin x + C$$

← $\int \frac{1}{x} dx = \ln |x| + C$

← $\int \cos x dx = \sin x + C$

• (+ و -)

$$f(x) = g'(x) \quad \text{No.}$$

Q:- what is the Anti-derivative of $f'(x)$.

$$\Rightarrow F(x) = \int f'(x) + c$$

$$\boxed{\int f'(x) \cdot dx = f(x) + c}, \quad \boxed{\int f''(x) \cdot dx = f'(x) + c}$$

$$1. \int k \cdot dx = kx + c$$

$$2. \int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

Power $\downarrow$ by 1

$$3. \int \frac{1}{x} \cdot dx = \log|x| + c$$

$$4. \int e^x \cdot dx = e^x + c$$

$$5. \int \frac{f'(x)}{f(x)} = \ln|f(x)| + c$$

$$6. \int \sin x \cdot dx = -\cos x + c$$

$$7. \int \cos x \cdot dx = \sin x + c$$

$$8. \int \sec^2 x \cdot dx = \tan x + c$$

$$9. \int \sec x \tan x \cdot dx = \sec x + c$$

$$10. \int \csc^2 x \cdot dx = -\cot x + C$$

$$11. \int \csc x \cot x \cdot dx = -\csc x + C$$

Rule :-

* بقسمة، ضرب، اشتراك لا توزع.

$$1. \int a f(x) \cdot dx = a \int f(x) \cdot dx$$

$$2. \int f(x) \pm g(x) \cdot dx = \int f(x) \pm \int g(x) \cdot dx$$

Ex. Evaluate :-

$$1. \int \sqrt{x} - \frac{3}{x^2} dx$$

$$= \int (x)^{1/2} - 3 \int x^{-2} \cdot dx$$

$$= \frac{x^{3/2}}{3/2} - 3 \cdot \frac{x^{-1}}{-1} + C$$

$$= \frac{2}{3} x^{3/2} + 3 \frac{1}{x} + C$$

$$= \frac{2}{3} \underline{\underline{x\sqrt{x}}} + 3 \frac{1}{x} + C$$

$$\sqrt{x^3}$$

$$\sqrt{x^2} \sqrt{x}$$

$$x\sqrt{x}$$

$$2. \int \frac{x^2 \sin x - \sqrt{x}}{x^2} dx$$

$$= \int \sin x - \frac{\sqrt{x}}{x^2} dx$$

بعد التقطع $\frac{1}{2} - 2 = -\frac{3}{2}$

$$= \int \sin x - x^{-3/2} dx$$

$$= -\cos x - \frac{x^{-1/2}}{-1/2} + C$$

$$= -\cos x + 2 \frac{1}{\sqrt{x}} + C$$

Rule number 5

$$3. \int \frac{2x+1}{x^2+x-3} dx$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

Trick Question:-

$$1. \int \frac{\sin(2x)}{\cos x} dx$$

$$* \sin(2x) = 2 \sin x \cos x$$

$$\int \frac{2 \sin x \cos x}{\cos x} dx$$

$$= 2 \int \sin x dx$$

$$= -2 \cos x + C$$

$$2. \int \tan^2 x \cdot dx$$

$$* \tan^2 x + 1 = \sec^2 x$$

$$= \int (\sec^2 x - 1) \cdot dx = \tan x - x + C$$

Ex. Find $f(x)$ so that $f'(x) = \frac{1}{x}$, $f(1) = 2$

$$= \int f'(x) = \int \frac{1}{x} \cdot dx$$

$$f(x) = \ln |x| + C$$

$$f(1) = \ln |1| + C$$

$$\underline{2=C} \Rightarrow * f(x) = \ln |x| + 2$$

* Indefinite megeral [Anti-derivative]

$$\int f'(x) \cdot dx = f(x) + c$$

$$\int f''(x) \cdot dx = f'(x) + c$$

Ex. Find $f(x)$ so that $f'(x) = 6x + 12$ $f(0) = 4$

$$f'(1) = 2$$

$$\int f''(x) \cdot dx = \int 6x + 12$$

$$\Rightarrow f'(x) = 3x^2 + 12x + A$$

$$* \int f'(1) = \int 3x^2 + 12x + A$$

$$\Rightarrow f(x) = x^3 + 6x^2 + Ax + B$$

$$* f'(1) = 2 \Rightarrow 2 = 3 + 12 + A \Rightarrow \boxed{A = -13}$$

$$f(0) = 4 = B$$

$$f(x) = x^3 + 6x^2 - 13x + 4$$

Ex. Evaluate :-

$$* 1. \int \tan x \cdot dx = \int \frac{\sin x}{\cos x} \cdot dx$$

$$= - \int \frac{-\sin x}{\cos x} \cdot dx \Rightarrow \left(\int \frac{f'(x)}{f(x)} \right)$$

$$= -\ln |\cos x| + c$$

$$2. \int \frac{e^x}{u + e^x} \cdot dx = \ln |u + e^x| + c$$

6.2

* Integration By Substitution :-

- $\int 2x (\sin x^2) \cdot dx$ * اكله لانه سهل *

- $\int (\ln x + 3)^4 \cdot dx$

- $\int (2x+1) e^{(x^2+x+3)} \cdot dx$

$\Rightarrow \int f(u) u'(x) \cdot dx = \int f(u) \cdot du$

↳ integration by substitution.

Ex. Evaluation :-

1. $\int 2x \sin x^2 \cdot dx$

~~$\frac{1}{2} \int 2x \cdot \sin(x^2) \cdot dx$~~

* نوع اول $u = x^2 \Rightarrow \frac{du}{dx} = 2x \Rightarrow du = 2x \cdot dx$

$= \frac{1}{2} \int \sin(u) \cdot du$

$= -\frac{1}{2} \cos(u) + C$

$= -\frac{1}{2} \cos(x^2) + C$

$$2. \int (2x+1) e^{(x^2+x+3)} \cdot dx$$

نشان $u = x^2 + x + 3$

$$\frac{du}{dx} = (2x+1) \Rightarrow du = (2x+1) \cdot dx$$

$$= \int e^u \cdot du = e^u + C = e^{(x^2+x+3)} + C$$

$$3. \int \frac{(\ln x + 3)^4}{x} \cdot dx$$

$$= \int \frac{1}{x} \cdot (\ln x + 3)^4 \cdot dx$$

نشان $u = \ln x + 3 \Rightarrow \frac{du}{dx} = \frac{1}{x} \cdot dx$

$$= \int u^4 \cdot du = \frac{u^5}{5} + C = \frac{(\ln x + 3)^5}{5} + C$$

$$Ex. \frac{1}{2} \int 2(2x+3)^4 x \cdot dx$$

$$u = 2x+3 \Rightarrow x = \frac{u-3}{2}$$

$$du = 2 \cdot dx$$

$$= \frac{1}{2} \int u^4 \left(\frac{u-3}{2} \right) \cdot du$$

$$= \frac{1}{4} \int u^5 - 3u^4 \cdot du$$

$$= \frac{1}{4} \left(\frac{u^6}{6} - \frac{3}{5} u^5 \right) + C$$

$$= \frac{1}{4} \left(\frac{(2x+3)^6}{6} - \frac{3}{5} ((2x+3)^5) \right) + C$$

$$\text{Ex. } \int e^{ax} \cdot dx = \frac{1}{a} \int e^u \cdot du$$

$$u = ax \Rightarrow du = a \cdot dx$$

$$\frac{1}{a} \int e^u \cdot du$$

$$\int e^{ax} \cdot dx = \frac{1}{a} e^{ax} + C$$

Rule:-

$$\int f(ax+b) dx = \frac{1}{a} f(ax+b) + C$$

Ex. Evaluate

$$\begin{aligned} 1. \int \sin(2x+3) \cdot dx \\ = -\frac{1}{2} \cos(2x+3) + C \end{aligned}$$

$$\begin{aligned} 2. \int (3x-1)^8 \cdot dx \\ = \frac{1}{3} \frac{(3x-1)^9}{9} + C \end{aligned}$$

$$\begin{aligned} 3. \int \frac{1}{2x-3} \cdot dx \\ = \frac{1}{2} \ln|2x-3| + C \end{aligned}$$

$$\Rightarrow \int \frac{1}{ax+b} \cdot dx = \frac{1}{a} \ln|ax+b| + C$$

* **Ex.** $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} \cdot dx$

$$2 \int \frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}} \cdot dx$$

نقطة $u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}} \cdot dx$

$$\Rightarrow 2 \int e^u \cdot du$$

$$= 2e^u + C$$

$$= 2e^{\sqrt{x}} + C$$

$$* \int \frac{P(x)}{Z(x)} \cdot dx, \text{ degree } P(x) < \text{degree } Z(x)$$

Ex.

$$\boxed{1} \int \frac{2x+4}{x^2+4x-8} dx$$

$$* \int \left(\frac{F'(x)}{F(x)} \right)$$

$$= \ln |x^2+4x-8| + C$$

$$\boxed{2} \int \frac{2x}{x^2+4x-5} \cdot dx$$

* دائمًا لنقسم البسط بـ x^2+4x-5 لأن من الجان.

$$\frac{2x}{x^2+4x-5} = \frac{A}{(x+5)} + \frac{B}{(x-1)}$$

$$2x = A(x-1) + B(x+5)$$

$$\text{at } x=1 \Rightarrow 2 = 6B \Rightarrow \boxed{B = 1/3}$$

$$\text{at } x=-5 \Rightarrow -10 = -6A \Rightarrow \boxed{A = 5/3} \quad * \text{ then check } \text{نقوم بالتحقق}$$

$$\int \frac{2x}{x^2+4x-5} \cdot dx = \int \frac{5/3}{x+5} + \frac{1/3}{x-1} \cdot dx$$

$$= 5/3 \ln |x+5| + 1/3 \ln |x-1| + C$$

القاعدة من التعريف

$$\int f(ax+b) \cdot dx = \frac{1}{a} \int f(x) \cdot dx + C$$

$$\text{Ex: 1. } \int e^{2x-1} \cdot dx = \frac{1}{2} e^{2x-1} + C$$

$$2. \int \sin(3x) \cdot dx = \frac{-\cos(3x)}{3} + C$$

$$3. \int \frac{a}{bx+k} \cdot dx = a \int \frac{1}{bx+k} \cdot dx = \frac{a}{b} \ln|bx+k| + C$$

$$4. \int \frac{1}{bx+k} \cdot dx = \frac{1}{b} \ln|bx+k| + C$$

* انتبه! اكبر منه بسط ✓ خذ على شكل

$$\boxed{3} \int \frac{x+3}{x^3-4x} \cdot dx$$

$$\int \frac{x+3}{x(x-2)(x+2)} \cdot dx$$

$$\int \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2} \cdot dx$$

$$\frac{x+3}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2}$$

$$= \frac{A(x-2)(x+2)}{x(x-2)(x+2)} + \frac{B(x+2)(x)}{x-2(x+2)(x)} + \frac{C(x-2)(x)}{x+2(x-2)(x)}$$

$$x+3 = A(x-2)(x+2) + Bx(x+2) + Cx(x-2)$$

$$\text{at } x=0 \Rightarrow 3 = -4A \Rightarrow \boxed{A = -3/4}$$

$$\text{at } x=2 \Rightarrow 5 = 8B \Rightarrow \boxed{B = 5/8}$$

$$\text{at } x=-2 \Rightarrow 1 = 8C \Rightarrow \boxed{C = 1/8}$$

$$\int \frac{-3/4}{x} + \frac{5/8}{x-2} + \frac{1/8}{x+2} \cdot dx$$

$$= \frac{-3}{4} \ln|x| + \frac{5}{8} \ln|x-2| + \frac{1}{8} \ln|x+2|$$

Ex. $\int \frac{x^2}{x^2-1} \cdot dx$ * انكسار بساط من صيغة (نقد)

method (1) :- $\int \frac{x^2-1}{x^2-1} + \frac{1}{x^2-1} \cdot dx$

$$\int 1 \cdot dx + \int \frac{1}{x^2-1} \cdot dx$$

$$\frac{1}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$= \frac{A(x+1) + B(x-1)}{(x+1)(x-1)}$$

$$1 = A(x+1) + B(x-1)$$

$$\text{at } x=1 \Rightarrow 1 = 2A \Rightarrow \boxed{A = 1/2}$$

$$\text{at } x=-1 \Rightarrow 1 = 2B \Rightarrow \boxed{B = 1/2}$$

مربع $\Rightarrow$

Ex. $\int \frac{\sqrt{x}}{x-a} \cdot dx$

نقشه $t = \sqrt{x} \Rightarrow dt = \frac{1}{2\sqrt{x}} \cdot dx$
 $2\sqrt{x} \cdot dt = dx$
 $2\sqrt{t} \cdot dt = dx$

$$\frac{\sqrt{x}}{x-a} = \int \frac{t}{t^2-a} \cdot 2t \cdot dt$$

$$\int \frac{2t^2}{t^2-a} \cdot dt$$

$$\int \frac{2(t^2-a)}{t^2-a} + \frac{18}{t^2-a} \cdot dt$$

$$\int 2 dt + \int \frac{18}{t^2-a} \cdot dt$$

$$\frac{18}{t^2-a} = \frac{A}{(t-3)} + \frac{B}{(t+3)}$$

$$= \frac{A(t+3) + B(t-3)}{(t-3)(t+3)}$$

$$18 = A(t+3) + B(t-3)$$

$$\text{at } t=3 \Rightarrow 18 = 6A \Rightarrow \boxed{A=3}$$

$$\text{at } t=-3 \Rightarrow 18 = -6B \Rightarrow \boxed{B=-3}$$

$$\int 2 dt + \int \frac{3}{t-3} dt + \int \frac{-3}{t+3} dt$$

$$= 2t + 3 \ln|t-3| - 3 \ln|t+3| + C$$

$$= 2\sqrt{x} + 3 \ln|\sqrt{x}-3| - 3 \ln|\sqrt{x}+3| + C$$

$\Rightarrow$

6.

* Integration by parts :-

$\int \text{Polynomial} * \left[\begin{array}{c} \sin x \\ e^x \\ \ln x \end{array} \right] dx$

Notes: $u(x)$ is the polynomial, $v(x)$ is the function in the bracket. $u'(x)$ is the derivative of $u(x)$.

$$\int u(x) \cdot v'(x) \cdot dx = u(x) \cdot v(x) - \int u'(x) \cdot v(x) \cdot dx.$$

1. اختيار واحد مماثل لـ $v'(x)$ والآخر $u(x)$.

Ex. [1] $\int x \sin x \cdot dx$

2. نأخذ المشتقة للـ $v(x)$ ونأخذ واحد من $u(x)$.

$u(x) = x \Rightarrow u'(x) = 1$ (ما قبل مشتقة)
 $v(x) = \sin x \Rightarrow v'(x) = \cos x$ (ما قبل مشتقة)

method [1]

$u(x) = x$ $v'(x) = \sin x$
 $u'(x) = 1$ $v(x) = -\cos x$

$$\int \sin x \cdot dx = -x \cos x + \int \cos x \cdot dx$$

$$= -x \cos x + \sin x + C$$

$$[-2 * -\cos]$$

method [2]

$u(x) = x$ $v'(x) = \sin x$
 $= 1$ $= -\cos x$
 $= 0$ $= -\sin x$

$$= -x \cos x + \sin x + C$$

$$\boxed{2} \int (x^2 - x + 1) e^x \cdot dx$$

method $\boxed{1}$

$$\begin{array}{ll} u(x) = x^2 - x + 1 & v'(x) = e^x \\ u'(x) = 2x - 1 & v(x) = e^x \end{array}$$

$$\int (x^2 - x + 1) e^x \cdot dx = (x^2 - x + 1) e^x - \int (2x - 1) e^x \cdot dx$$

$$\begin{array}{ll} u(x) = 2x - 1 & v'(x) = e^x \\ u'(x) = 2 & v(x) = e^x \end{array}$$

$$\int (x^2 - x + 1) e^x \cdot dx = (x^2 - x + 1) e^x - (2x - 1) e^x - \int 2e^x \cdot dx$$

$$\int (x^2 - x + 1) e^x \cdot dx = (x^2 - x + 1) e^x - (2x - 1) e^x + \underline{2e^x + C}$$

methode $\boxed{2}$

$$\begin{array}{ll} u(x) = x^2 - x + 1 & v'(x) = e^x \\ = 2x - 1 & \rightarrow = e^x \\ = 2 & \rightarrow = e^x \\ = 0 & \rightarrow = e^x \end{array}$$

$$= (x^2 + x - 1) e^x - (2x - 1) e^x + \cancel{2x} 2e^x + C$$

$$\boxed{3} \int x^3 \cdot \ln x \cdot dx$$

$$u(x) = \ln x$$

$$v(x) = x^3$$

$$u'(x) = \frac{1}{x}$$

$$v'(x) = \frac{x^4}{4}$$

$$\begin{aligned} &= (\ln x) \left(\frac{x^4}{4} \right) - \int \frac{x^4}{4} \cdot \frac{1}{x} \cdot dx \\ &= \ln x \frac{x^4}{4} - \frac{1}{4} \int x^3 \cdot dx \\ &= \frac{x^4}{4} \ln x - \frac{1}{4} \cdot \frac{x^4}{4} + C \end{aligned}$$

$$\boxed{4} \int (3x-1)(x-2)^{1/2} \cdot dx$$

method [1] : Substitution :-

$$t = x-2 \Rightarrow dt = dx$$

$$t+2 = x$$

$$\begin{aligned} &= \int (3t+5) + t^{1/2} \cdot dt \\ &= \int 3t^{3/2} + 5t^{1/2} \cdot dt \end{aligned}$$

$$= \frac{3t^{5/2}}{5/2} + \frac{5t^{3/2}}{3/2} + C$$

$$= \frac{3(x-2)^{5/2}}{5/2} + \frac{5(x-2)^{3/2}}{3/2} + C$$

methode [2] : by parts :-

Date. No.

$$u(x) = 3x - 1 \quad v'(x) = (x-2)^{1/2}$$

$$u'(x) = 3 \quad v(x) = \frac{2(x-2)^{3/2}}{3}$$

$$= \frac{2(3x-1)(x-2)^{3/2}}{3} - \int 2(x-2)^{3/2} \cdot dx$$

$$= \frac{2}{3} (3x-1)(x-2)^{3/2} - \frac{2(x-2)^{5/2}}{2/5} + C$$

* اگر طریقہ جواب بنانے مختلف بسبب اختلاف فوراً ہی پس خاوی
عنانہ شاکیہ نسخہ جواب اپنی پر مبنی اصل

Chapter 7 :- Definite integral

Date.

No.

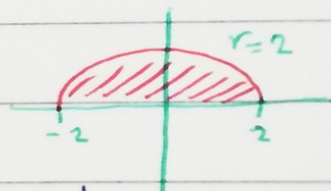
Def:- let $f(x)$ be defined on $[a, b]$:-

1. if $f(x) \geq 0$, then $\int_a^b f(x) \cdot dx = A = \text{the area under } f(x)$

2. if $f(x) \leq 0$, then $\int_a^b f(x) \cdot dx = -A$

Ex. Find the integral $\int_{-2}^2 f(x) \cdot dx$ where $f(x) = \sqrt{4-x^2}$

$$\int_{-2}^2 \sqrt{4-x^2} \cdot dx = \frac{r^2 \pi}{2} = \frac{(2)^2 \pi}{2} = 2\pi$$



Q:- Is there other method to find $\int_a^b f(x) \cdot dx$?

$\Rightarrow$ yes, we use the indefinite integral

Theorem: (The fundamental theorem of calculus I)

$$\int_a^b f(x) \cdot dx = F(x) \Big|_a^b = F(b) - F(a)$$

Ex. Evaluate:-

$$1. \int_0^1 x^3 \cdot dx = \left[\frac{x^4}{4} \right]_0^1 = F(1) - F(0) = \frac{(1)^4}{4} - \frac{(0)^4}{4} = \frac{1}{4}$$

$$2. \int_0^\pi \sin\left(\frac{x}{2}\right) \cdot dx = \left[\frac{-\cos\left(\frac{x}{2}\right)}{\frac{1}{2}} \right]_0^\pi = F(\pi) - F(0)$$

$$= \frac{-\cos\left(\frac{\pi}{2}\right)}{\frac{1}{2}} - \frac{-\cos 0}{\frac{1}{2}} = \boxed{2}$$

$$3. \int_e^{e^2} \frac{1}{x \ln x} \cdot dx$$

دوره وضع کردن

method (1):- $\int \frac{1}{x} \cdot \frac{1}{\ln x} \cdot dx$

$$t = \ln x \Rightarrow dt = \frac{1}{x} \cdot dx$$

$$= \int \frac{1}{t} \cdot dt = \ln|t| + C$$

$$F(x) = \ln|\ln x| + C$$

$$= \left[\ln|\ln(x)| \right]_e^{e^2} = \ln|\ln e^2| - \ln|\ln e|$$

$$= \ln(2) - \ln 1$$

$$= \ln 2$$

* بی نهایت

methode (2):- $\int_e^{e^2} \frac{1}{x} \cdot \frac{1}{\ln x} \cdot dx$

$$t = \ln x \Rightarrow dt = \frac{1}{x} \cdot dx$$

$$\int_1^2 \frac{1}{t} \cdot dt$$

$$* dt \begin{cases} \text{as } x=e \\ t=1 \\ \text{as } x=e^2 \\ t=2 \end{cases}$$

$$= \left[\ln|t| \right]_1^2 = \ln(2) - \cancel{\ln(1)} = \ln 2$$

* properties of definite integral

[v.I]

$$1. \int_a^b k \cdot dx = (b-a)k$$

$$2. \int_a^a f(x) \cdot dx = 0$$

$$3. \int_a^b f(x) \cdot dx = - \int_b^a f(x) \cdot dx$$

$$4. \int_a^b f(x) \cdot dx = \int_a^c f(x) \cdot dx + \int_c^b f(x) \cdot dx$$

Ex. Evaluate let $\int_0^2 f(x) \cdot dx = 4$, $\int_0^2 f(-x) \cdot dx = 8$
and $\int_0^2 3g(x) \cdot dx = 6$, Find :-

$$1. \int_0^2 4(f(x)) - \underline{g(x)} \cdot dx$$

g is given
minus 2/3
2

$$= 4 \int_0^2 f(x) \cdot dx - \int_0^2 g(x) \cdot dx$$

$$= 4(4) - 2 = 14$$

$$2. \int_2^2 g(x) \cdot dx = 0$$

$$\begin{aligned} 3. \int_0^2 x^2 + g(x) \cdot dx &= \int_0^2 x^2 \cdot dx + \int_0^2 g(x) \cdot dx \\ &= \left[\frac{x^3}{3} \right]_0^2 + 2 \\ &= \frac{8}{3} + 2 = \frac{14}{3} \end{aligned}$$

$$4. \int_{-2}^2 f(x) dx = \int_{-2}^0 f(x) dx + \int_0^2 f(x) dx = 4 = \boxed{12}$$

تقسيم إلى جزئين

منه معلوم أن $8 =$

$$\text{but } 8 = \int_{-2}^2 f(-x) dx = \int_0^{-2} f(t) dt = \int_{-2}^0 f(t) dt$$

$$\begin{aligned} t &= -x & \begin{cases} \text{as } x=0 \rightarrow t=0 \\ \text{as } x=2 \rightarrow t=-2 \end{cases} \\ dt &= -dx \end{aligned}$$

Ex. Find $\frac{d}{dx} \int_0^2 \sin^2(2t) dt = 0$

Theorem: The fundamental of calculus II)

let $f(x)$ be continuous on $[a, b]$ and $g(x) = \int_a^x f(t) dt$

Then $g'(x) = f(x)$

$$* \left| \frac{d}{dx} \int_a^x f(t) dt = f(x) \right|$$

$$* \left| \frac{d}{dx} \int_a^{u(x)} f(t) dt = f(u(x)) \cdot u'(x) \right|$$

Ex. let $g(x) = \int_x^2 \frac{1}{1+t^2} dt = - \int_3^x \frac{1}{1+t^2} dt$, find $g'(2)$

$$\begin{aligned} g'(x) &= -\frac{1}{1+x^2} \\ g'(2) &= -\frac{1}{5} \end{aligned}$$

Ex. let $g(x) = \sin \int_0^{3x^2} \cos(t^2) \cdot dt$ find $g'(x)$

$$g'(x) = \sin \cdot \cos(9x^4) \cdot 6x + \cos x \int_0^{3x^2} \cos(t^2) \cdot dt$$

Ex. find $\frac{d}{dx} \int_x^{e^x} \tan(t) \cdot dt$?

$$g(x) = \int_x^{e^x} \tan(t) \cdot dt = - \int_a^x \tan t \cdot dt + \int_a^{e^x} \tan t \cdot dt$$

$$g'(x) = -\tan x + \tan(e^x) \cdot e^x$$

Ex. $f(x) = \begin{cases} x^2 & , 0 \leq x \leq 1 \\ \sqrt{x} & , 1 < x \leq 4 \end{cases}$ find $\int_0^4 f(x) \cdot dx$

$$= \int_0^1 f(x) \cdot dx + \int_1^4 f(x) \cdot dx$$

$$= \int_0^1 x^2 \cdot dx + \int_1^4 \sqrt{x} \cdot dx$$

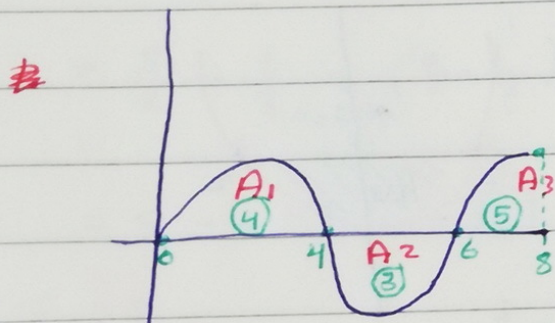
$$= \frac{x^3}{3} \Big|_0^1 + \frac{2}{3} \cdot x^{3/2} \Big|_1^4$$

$$= \left(\frac{1}{3} - 0\right) + \frac{16}{3} - \frac{2}{3}$$

$$= \frac{15}{3} = \boxed{5}$$

EXE. Evaluate $\int_0^4 |x-2| \cdot dx$

EX. The following shows the graph of $y=f(x)$



1. Area ذاتياً + اورو
 2. Integral كادى كيونى سالو
 حسب موقع وصوره اراخت
 ولا موفه

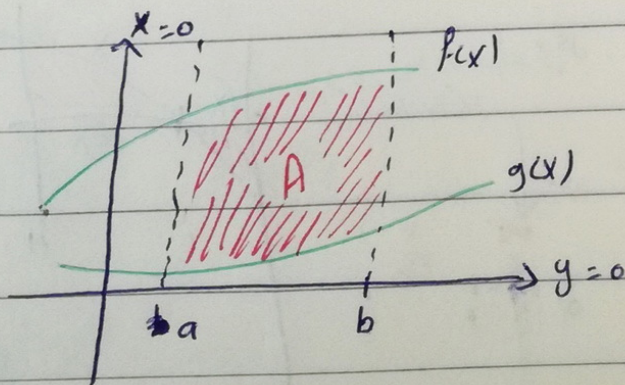
$$1. \int_0^8 f(x) \cdot dx = \int_0^4 f(x) \cdot dx + \int_4^6 f(x) \cdot dx + \int_6^8 f(x) \cdot dx$$

$$= (4) + (-3) + 5 = \boxed{6}$$

$$2. \int_0^8 |f(x)| \cdot dx = 4 + 3 + 5 = \boxed{12}$$

Rule:- let $f(x) \geq g(x)$ on $[a, b]$ فرضان ادمها x و f و g

then the $A = \int_a^b f(x) - g(x) \cdot dx$ دى مدينه نقيده مدينه



د كا كادى د مدينه شامو ميا كادى
 د ف و گ شى مدينه مدينه
 شامو مدينه

Ex. Find the area bounded by $f(x) = x^2 - 4$ and the x -axis. → $g(x) = 0$

$$f(x) = g(x)$$

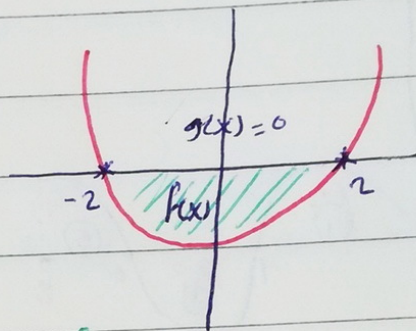
$$x^2 - 4 = 0 \Leftrightarrow x = 2, -2$$

$$A = \int_{-2}^2 0 - (x^2 - 4) \cdot dx$$

$$= \int_{-2}^2 -x^2 + 4 \cdot dx = \left[-\frac{x^3}{3} + 4x \right]_{-2}^2$$

$$= \left(-\frac{8}{3} + 8 \right) - \left(\frac{8}{3} - 8 \right)$$

$$\frac{-16}{3} + \frac{16 \cdot 3}{3} = \frac{32}{3} \rightarrow \text{must be positive}$$



سوال دیگر

Ex. Find the area bounded by $y = \sqrt{x}$, $y = x$, $x = 0$, $x = 4$

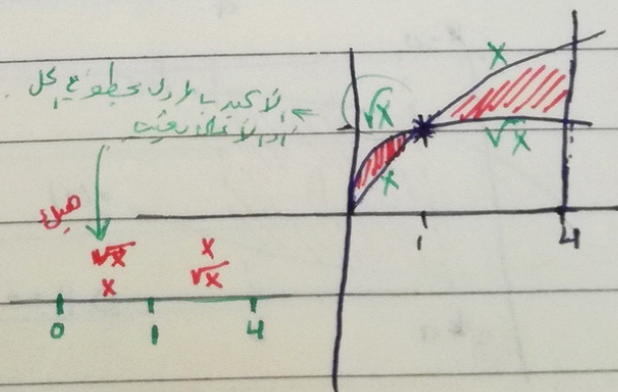
$g(x)$ $f(x)$ $h(x)$ $k(x)$

$$f(x) = \sqrt{x}, g(x) = x$$

$$f(x) = g(x)$$

$$\sqrt{x} = x \rightarrow \text{بعضی نقاط با هم تداخل دارند}$$

$$x^2 - x = 0 \Rightarrow x = 0, 1$$



مساحت بین دو تابع (ایکدیگر)

Date.

No.

$$A = \int_0^1 \sqrt{x} - x \cdot dx + \int_1^4 x - \sqrt{x} \cdot dx$$

$$A = \frac{2}{3} x^{3/2} \Big|_0^1 - \frac{x^2}{2} \Big|_0^1 + \frac{x^2}{2} \Big|_1^4 - \frac{2}{3} x^{3/2} \Big|_1^4$$

$$= \frac{3}{2} - \frac{1}{2} + \left(8 - \frac{1}{2}\right) - \left(\frac{16}{3} - \frac{2}{3}\right)$$

$$= 8 - 4 - 1 = \boxed{3}$$

Trick Q.

Ex. if the area bounded by $f(x) = \sqrt{x}$, x -axis from $x=0$ to $x=a > 0$ is $\frac{16}{3}$ units. find a .

$$A = \frac{16}{3}$$

$$\int_0^a \sqrt{x} - 0 \cdot dx = \frac{16}{3}$$

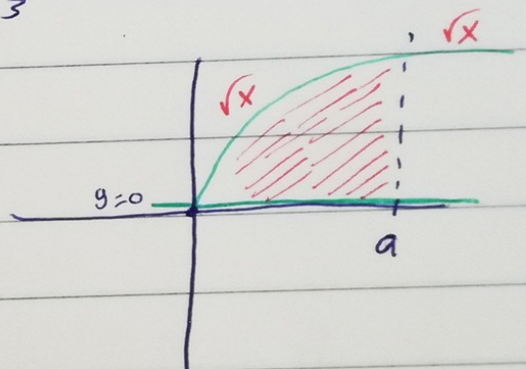
$$\frac{2}{3} x^{3/2} \Big|_0^a = \frac{16}{3}$$

$$\frac{2}{3} a^{3/2} = \frac{16}{3}$$

$$(a^{3/2})^{1/2} = (8)^{1/3}$$

$$(a^{1/2})^2 = (2)^2$$

$$\boxed{a=4}$$



* Function میں سے لے کر
 دے دے کہہ سکتے ہیں